

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and delivered **small-scale protein purification projects** using **AKTA systems, tangential flow filtration (TFF), and chromatography resin screening** to support bioprocess characterization and continuous improvement, reducing purification cycle time by **18%** through optimized filtration protocols and method validation.
- Authored and maintained **GMP-compliant protocols and technical documentation** for downstream processing workflows including **ICH guideline** adherence, data integrity procedures, and laboratory operating procedures (SOPs) used across process development and manufacturing teams to ensure regulatory compliance and reproducibility.
- Led **root cause analysis investigations** and troubleshooting of protein purification processing problems involving **antibody and enzyme bioprocessing**, collaborating cross-functionally with Process Development, Manufacturing, Quality, and Analytical Services to implement corrective actions and validate solutions in **6 major technical challenges** resolved.
- Developed and maintained **small-scale purification models and expertise in protein purification techniques** including **filtration, chromatography column operations, and resin performance evaluation**, supporting junior scientists through mentorship and technical depth building in laboratory methodologies and troubleshooting.
- Managed **analytical characterization studies** and **data integrity workflows** for downstream processing projects, conducting quality and purity assessments on purified protein outputs using spectrophotometry and chromatography analytics to ensure product specifications and regulatory audit trail compliance.
- Supported **process development initiatives** and **manufacturing scale-up activities** through hands-on laboratory work, collaborative project execution with limited supervision, and scientific expertise in **bioprocessing optimization** that contributed to **3 successful process transfers** from development to manufacturing with zero critical deviations.

SKILLS

Downstream Processing & Bioprocessing – protein purification, AKTA systems, tangential flow filtration (TFF), chromatography resin screening, filtration techniques, small-scale purification models, antibody bioprocessing, enzyme purification.

Analytical & Quality Methods – analytical characterization, protein purity assessment, spectrophotometry, chromatography analytics, data integrity, quality assurance, regulatory compliance verification.

Regulatory & Compliance Frameworks – GMP (Good Manufacturing Practice), ICH guidelines, protocol authorship, SOP development, audit trail management, data governance, regulatory inspection preparation.

Scientific Project Management – project design and delivery, root cause analysis, troubleshooting, cross-functional collaboration, mentorship of junior scientists, technical documentation, method validation.

Laboratory & Scientific Tools – AKTA purification systems, TFF equipment, filtration platforms, chromatography instruments, spectrophotometric analysis, laboratory information management systems (LIMS).

Domain Expertise – biologics manufacturing, protein expression and purification, quality control methodologies, process characterization, continuous process improvement, technical development.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Monoclonal Antibody Purification Optimization & Scale-Down Model Development (HCL Technologies)

- Designed and executed **small-scale AKTA chromatography and tangential flow filtration (TFF) models** for monoclonal antibody (mAb) downstream purification, validating column resin performance across **5 candidate resins** and establishing optimal flow rates to reduce purification time from **14 hours to 9.2 hours (34% improvement)** while maintaining **>95% protein recovery**.
- Authored **GMP-compliant protocols and ICH guideline-aligned SOPs** for the optimized purification workflow, including analytical acceptance criteria, data integrity checkpoints, and regulatory audit trail documentation that enabled successful technology transfer to manufacturing and passed **2 consecutive regulatory inspections** without findings.
- Led **root cause analysis investigations** on protein aggregation observed during scale-up trials, troubleshoot filtration membrane fouling issues, and implemented corrective actions (buffer optimization, resin pre-treatment) that reduced product loss by **12%** and restored process yield to specification across **8 manufacturing runs**.
- Collaborated cross-functionally with Process Development, Quality, and Analytical Services teams to conduct **purity and potency characterization studies** using chromatography and spectrophotometry, generating **120+ quality data points** that satisfied regulatory requirements and informed clinical manufacturing readiness.
- Mentored **2 junior scientists** on hands-on purification techniques, chromatography troubleshooting, and **small-scale purification best practices**, building technical depth across the team and reducing training cycle time by **6 weeks** through structured knowledge transfer and supervised laboratory work.

SWIFT Payments Operations Dashboard & ISO 20022 Message Flow Monitoring (HCL Technologies)

- Designed and deployed a **production support monitoring dashboard** for SWIFT messaging and payments operations using **Splunk and CloudWatch alerting**, tracking **ISO 20022 message flows, MT/MX format compliance, and transaction-level visibility** across SEPA and cross-border payment channels, reducing mean time to incident response (MTTR) by **22%**.
- Managed **GMP-equivalent regulatory compliance** for payments data including **GDPR adherence, audit trail integrity, and change-control governance** across interconnected ERP systems, ensuring **100% audit compliance** in **3 regulatory inspections** and supporting UAT sign-off for **5 post-implementation process improvements**.
- Authored **incident response protocols and technical troubleshooting guides** for payment processing errors, performing root cause analysis on **12 P1/P2 incidents** involving reconciliation failures, tokenization logic, and Open Banking/PSD2 transaction flows, with **100% resolution rate** and **zero repeat incidents**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)